

Coverage test

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== tests coverage =====
----- coverage: platform win32, python 3.12.0-final-0 -----

Name      Stmts   Miss  Cover   Missing
-----
app.py      242     32    87%    74, 120, 127, 156, 185-187, 318, 440, 563-564, 590-643, 652-657
-----
TOTAL       242     32    87%
===== 53 passed, 1084 warnings in 18.64s =====
PS C:\Project\flask_auction>
```

Functional test

```
PS C:\Project\flask_auction> pytest test_app.py::FunctionalTestCase -v
===== test session starts =====
platform win32 -- Python 3.12.0, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\sa790\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Project\flask_auction
plugins: cov-7.0.0
collected 29 items

test_app.py::FunctionalTestCase::test_auction_no_winner_if_no_bids PASSED [ 3%]
test_app.py::FunctionalTestCase::test_auction_winner_selection PASSED [ 6%]
test_app.py::FunctionalTestCase::test_bid_creates_audit_trail PASSED [ 10%]
test_app.py::FunctionalTestCase::test_bid_increments_correctly PASSED [ 13%]
test_app.py::FunctionalTestCase::test_create_item_invalid_bid PASSED [ 17%]
test_app.py::FunctionalTestCase::test_create_item_invalid_duration PASSED [ 20%]
test_app.py::FunctionalTestCase::test_create_item_requires_login PASSED [ 24%]
test_app.py::FunctionalTestCase::test_create_item_short_description PASSED [ 27%]
test_app.py::FunctionalTestCase::test_create_item_short_name PASSED [ 31%]
test_app.py::FunctionalTestCase::test_create_item_success PASSED [ 34%]
test_app.py::FunctionalTestCase::test_place_bid_equal_to_current PASSED [ 37%]
test_app.py::FunctionalTestCase::test_place_bid_invalid_amount PASSED [ 41%]
test_app.py::FunctionalTestCase::test_place_bid_lower_than_current PASSED [ 44%]
test_app.py::FunctionalTestCase::test_place_bid_on_ended_auction PASSED [ 48%]
test_app.py::FunctionalTestCase::test_place_bid_requires_login PASSED [ 51%]
test_app.py::FunctionalTestCase::test_place_bid_success PASSED [ 55%]
test_app.py::FunctionalTestCase::test_user_can_view_item_detail PASSED [ 58%]
test_app.py::FunctionalTestCase::test_user_can_view_own_auctions PASSED [ 62%]
test_app.py::FunctionalTestCase::test_user_dashboard_requires_login PASSED [ 65%]
test_app.py::FunctionalTestCase::test_user_login_invalid_password PASSED [ 68%]
test_app.py::FunctionalTestCase::test_user_login_invalid_username PASSED [ 72%]
test_app.py::FunctionalTestCase::test_user_login_success PASSED [ 75%]
test_app.py::FunctionalTestCase::test_user_logout PASSED [ 79%]
test_app.py::FunctionalTestCase::test_user_registration_duplicate_username PASSED [ 82%]
test_app.py::FunctionalTestCase::test_user_registration_password_mismatch PASSED [ 86%]
test_app.py::FunctionalTestCase::test_user_registration_short_password PASSED [ 89%]
test_app.py::FunctionalTestCase::test_user_registration_short_username PASSED [ 93%]
test_app.py::FunctionalTestCase::test_user_registration_success PASSED [ 96%]
test_app.py::FunctionalTestCase::test_view_nonexistent_item_returns_404 PASSED [100%]
```

Integration test

```
PS C:\Project\flask_auction> pytest test_app.py::IntegrationTestCase -v
===== test session starts =====
platform win32 -- Python 3.12.0, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\sa790\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Project\flask_auction
plugins: cov-7.0.0
collected 6 items

test_app.py::IntegrationTestCase::test_complete_auction_workflow PASSED [ 16%]
test_app.py::IntegrationTestCase::test_database_transaction_integrity PASSED [ 33%]
test_app.py::IntegrationTestCase::test_multiple_users_bidding_concurrently PASSED [ 50%]
test_app.py::IntegrationTestCase::test_password_hashing_integration PASSED [ 66%]
test_app.py::IntegrationTestCase::test_session_management_integration PASSED [ 83%]
test_app.py::IntegrationTestCase::test_user_can_view_bid_history PASSED [100%]
```

Security

```
PS C:\Project\flask_auction> pytest test_app.py::SecurityTestCase -v
===== test session starts =====
platform win32 -- Python 3.12.0, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\sa790\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Project\flask_auction
plugins: cov-7.0.0
collected 18 items

test_app.py::SecurityTestCase::test_bid_amount_precision_preserved PASSED [ 5%]
test_app.py::SecurityTestCase::test_bid_on_nonexistent_item_returns_404 PASSED [ 11%]
test_app.py::SecurityTestCase::test_cannot_access_other_users_data_via_id_manipulation PASSED [ 16%]
test_app.py::SecurityTestCase::test_cannot_bid_on_own_item PASSED [ 22%]
test_app.py::SecurityTestCase::test_cannot_enumerate_users_via_registration PASSED [ 27%]
test_app.py::SecurityTestCase::test_cannot_modify_bid_amount_after_placement PASSED [ 33%]
test_app.py::SecurityTestCase::test_invalid_item_id_returns_404 PASSED [ 38%]
test_app.py::SecurityTestCase::test_logout_clears_session PASSED [ 44%]
test_app.py::SecurityTestCase::test_negative_bid_amount_rejected PASSED [ 50%]
test_app.py::SecurityTestCase::test_password_hashing_uses_proper_algorithm PASSED [ 55%]
test_app.py::SecurityTestCase::test_password_verification_is_case_sensitive PASSED [ 61%]
test_app.py::SecurityTestCase::test_passwords_are_hashed_not_stored_plaintext PASSED [ 66%]
test_app.py::SecurityTestCase::test_post_to_protected_routes_requires_auth PASSED [ 72%]
test_app.py::SecurityTestCase::test_protected_routes_require_authentication PASSED [ 77%]
test_app.py::SecurityTestCase::test_seller_cannot_manually_select_winner PASSED [ 83%]
test_app.py::SecurityTestCase::test_session_contains_user_id PASSED [ 88%]
test_app.py::SecurityTestCase::test_sql_injection_prevention_username PASSED [ 94%]
test_app.py::SecurityTestCase::test_zero_bid_amount_rejected PASSED [100%]
```

Verification

```
PS C:\Project\flask_auction> python -m pytest verification.test.py -v
===== test session starts =====
platform win32 -- Python 3.12.0, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\sa790\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Project\flask_auction
plugins: cov-7.0.0
collected 76 items

verification.test.py::FunctionalTestCase::test_auction_no_winner_if_no_bids PASSED [ 1%]
verification.test.py::FunctionalTestCase::test_auction_winner_selection PASSED [ 2%]
verification.test.py::FunctionalTestCase::test_bid_creates_audit_trail PASSED [ 3%]
verification.test.py::FunctionalTestCase::test_bid_increments_correctly PASSED [ 5%]
verification.test.py::FunctionalTestCase::test_create_item_invalid_bid PASSED [ 6%]
verification.test.py::FunctionalTestCase::test_create_item_invalid_duration PASSED [ 7%]
verification.test.py::FunctionalTestCase::test_create_item_requires_login PASSED [ 9%]
verification.test.py::FunctionalTestCase::test_create_item_short_description PASSED [10%]
verification.test.py::FunctionalTestCase::test_create_item_short_name PASSED [11%]
verification.test.py::FunctionalTestCase::test_create_item_success PASSED [13%]
verification.test.py::FunctionalTestCase::test_place_bid_equal_to_current PASSED [14%]
verification.test.py::FunctionalTestCase::test_place_bid_invalid_amount PASSED [15%]
verification.test.py::FunctionalTestCase::test_place_bid_lower_than_current PASSED [17%]
verification.test.py::FunctionalTestCase::test_place_bid_on_ended_auction PASSED [18%]
verification.test.py::FunctionalTestCase::test_place_bid_requires_login PASSED [19%]
verification.test.py::FunctionalTestCase::test_place_bid_success PASSED [21%]
verification.test.py::FunctionalTestCase::test_user_can_view_item_detail PASSED [22%]
verification.test.py::FunctionalTestCase::test_user_can_view_own_auctions PASSED [23%]
verification.test.py::FunctionalTestCase::test_user_dashboard_requires_login PASSED [25%]
verification.test.py::FunctionalTestCase::test_user_login_invalid_password PASSED [26%]
verification.test.py::FunctionalTestCase::test_user_login_invalid_username PASSED [27%]
verification.test.py::FunctionalTestCase::test_user_login_success PASSED [28%]
verification.test.py::FunctionalTestCase::test_user_logout PASSED [30%]
verification.test.py::FunctionalTestCase::test_user_registration_duplicate_username PASSED [31%]
verification.test.py::FunctionalTestCase::test_user_registration_password_mismatch PASSED [32%]
verification.test.py::FunctionalTestCase::test_user_registration_short_password PASSED [34%]
verification.test.py::FunctionalTestCase::test_user_registration_short_username PASSED [35%]
verification.test.py::FunctionalTestCase::test_user_registration_success PASSED [36%]
verification.test.py::FunctionalTestCase::test_view_nonexistent_item_returns_404 PASSED [38%]
verification.test.py::IntegrationTestCase::test_complete_auction_workflow PASSED [39%]
verification.test.py::IntegrationTestCase::test_database_transaction_integrity PASSED [40%]
verification.test.py::IntegrationTestCase::test_multiple_users_bidding_concurrently PASSED [42%]
verification.test.py::IntegrationTestCase::test_password_hashing_integration PASSED [43%]
verification.test.py::IntegrationTestCase::test_session_management_integration PASSED [44%]
verification.test.py::IntegrationTestCase::test_user_can_view_bid_history PASSED [46%]
verification.test.py::SecurityTestCase::test_bid_amount_precision_preserved PASSED [47%]
verification.test.py::SecurityTestCase::test_bid_on_nonexistent_item_returns_404 PASSED [48%]
verification.test.py::SecurityTestCase::test_cannot_access_other_users_data_via_id_manipulation PASSED [50%]
verification.test.py::SecurityTestCase::test_cannot_bid_on_own_item PASSED [51%]
verification.test.py::SecurityTestCase::test_cannot_enumerate_users_via_registration PASSED [52%]
verification.test.py::SecurityTestCase::test_cannot_modify_bid_amount_after_placement PASSED [53%]
verification.test.py::SecurityTestCase::test_invalid_item_id_returns_404 PASSED [55%]
verification.test.py::SecurityTestCase::test_logout_clears_session PASSED [56%]
verification.test.py::SecurityTestCase::test_negative_bid_amount_rejected PASSED [57%]
verification.test.py::SecurityTestCase::test_password_hashing_uses_proper_algorithm PASSED [59%]
verification.test.py::SecurityTestCase::test_password_verification_is_case_sensitive PASSED [60%]
```

verification_test.py::SecurityTestCase::test_passwords_are_hashed_not_stored_plaintext PASSED	[61%
verification_test.py::SecurityTestCase::test_post_to_protected_routes_requires_auth PASSED	[63%
verification_test.py::SecurityTestCase::test_protected_routes_require_authentication PASSED	[64%
verification_test.py::SecurityTestCase::test_seller_cannot_manually_select_winner PASSED	[65%
verification_test.py::SecurityTestCase::test_session_contains_user_id PASSED	[67%
verification_test.py::SecurityTestCase::test_sql_injection_prevention_username PASSED	[68%
verification_test.py::SecurityTestCase::test_zero_bid_amount_rejected PASSED	[69%
verification_test.py::VerificationTestCase::test_all_tests_have_docstrings PASSED	[71%
verification_test.py::VerificationTestCase::test_authentication_coverage PASSED	[72%
verification_test.py::VerificationTestCase::test_base_test_case_helpers PASSED	[73%
verification_test.py::VerificationTestCase::test_base_test_case_setup PASSED	[75%
verification_test.py::VerificationTestCase::test_bidding_coverage PASSED	[76%
verification_test.py::VerificationTestCase::test_functional_test_class_exists PASSED	[77%
verification_test.py::VerificationTestCase::test_functional_test_class_inherits_base PASSED	[78%
verification_test.py::VerificationTestCase::test_functional_test_count PASSED	[80%
verification_test.py::VerificationTestCase::test_integration_test_class_exists PASSED	[81%
verification_test.py::VerificationTestCase::test_integration_test_class_inherits_base PASSED	[82%
verification_test.py::VerificationTestCase::test_integration_test_count PASSED	[84%
verification_test.py::VerificationTestCase::test_integration_transaction_testing PASSED	[85%
verification_test.py::VerificationTestCase::test_integration_workflow_testing PASSED	[86%
verification_test.py::VerificationTestCase::test_item_creation_coverage PASSED	[88%
verification_test.py::VerificationTestCase::test_security_authentication_coverage PASSED	[89%
verification_test.py::VerificationTestCase::test_security_password_coverage PASSED	[90%
verification_test.py::VerificationTestCase::test_security_session_coverage PASSED	[92%
verification_test.py::VerificationTestCase::test_security_sql_injection_coverage PASSED	[93%
verification_test.py::VerificationTestCase::test_security_test_class_exists PASSED	[94%
verification_test.py::VerificationTestCase::test_security_test_class_inherits_base PASSED	[96%
verification_test.py::VerificationTestCase::test_security_test_count PASSED	[97%
verification_test.py::VerificationTestCase::test_test_naming_convention PASSED	[98%
verification_test.py::VerificationTestCase::test_test_suite_total_count PASSED	[100%